Surd rules

1

a False **b** True

2

a $\sqrt{252}$ **b** $\sqrt{224}$ **c** $\sqrt{45}$ **d** $\sqrt{1000}$

3

a $4+3\sqrt{5}$ **b** $5\sqrt{2}$ **c** 45 **d** 735

e $6\sqrt{x}$ **f** $\sqrt{3}$ **g** $\frac{\sqrt{23}}{4}$

Multiplication and division of surds

4

a $\sqrt{22}$ **b** $4\sqrt{2}$ **c** $20\sqrt{7}$ **d** $15\sqrt{35}$

e $\sqrt{70}$ **f** $35\sqrt{21}$ **g** $\sqrt{35}$ **h** $80\sqrt{5}$

i $\sqrt{231}$ **j** $11\sqrt{5}$ **k** $20\sqrt{11}$ **l** $30\sqrt{55}$

m $14\sqrt{11}$ **n** $24\sqrt{15}$ **o** $320\sqrt{3}$ **p** $40\sqrt{51}$

q $340\sqrt{7}$ **r** $216\sqrt{17}$

5

a $\sqrt{5}$ **b** $2\sqrt{7}$ **c** 32 **d** $8\sqrt{11}$

e $4\sqrt{7}$ **f** $\sqrt{3}$ **g** $\sqrt{11}$ **h** $\sqrt{3}$

i $\sqrt{7}$ **j** $\sqrt{13}$ **k** $5\sqrt{7}$ **l** $10\sqrt{5}$

m $15\sqrt{2}$ **n** $2\sqrt{7}$ **o** 3 **p** 6

q 10 **r** 16 **s** 10 **t** $\frac{5}{9}$

u $\frac{9}{2}$

6

a 2 **b** $\frac{1}{\sqrt{5}}$ **c** 4 **d** $\frac{1}{\sqrt{3}}$

e $\frac{1}{\sqrt{3}}$ **f** 2 **g** $\frac{2}{3}$ **h** $\frac{3}{2}$

i $17\sqrt{7}$

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Mixed operations with surds

7

a $26\sqrt{3}$ b $-18\sqrt{5}$ c $27\sqrt{6}+11\sqrt{13}$ d $20\sqrt{6}+5\sqrt{11}$

e $4\sqrt{6}$ f $16\sqrt{5}$ g $-3\sqrt{2}$ h $42\sqrt{5}$

8

a $3\sqrt{3}$ b $3\sqrt{3}$ c $5\sqrt{2}$ d $6\sqrt{5}$

9

a $-5\sqrt{7}+\sqrt{35}$

b $7\sqrt{14}+42\sqrt{2}$

c $-42\sqrt{22}+7\sqrt{77}$

d $\sqrt{26}-3\sqrt{13}+5\sqrt{2}-15$

e $\sqrt{14}-\sqrt{77}-\sqrt{26}+\sqrt{143}$

f $12+2\sqrt{11}$

g $29+12\sqrt{5}$

h $\sqrt{77}+4\sqrt{11}$

i $3\sqrt{7}+\sqrt{21}$

j $-6\sqrt{2}+\sqrt{22}$

k $3\sqrt{39}-15\sqrt{3}$

l $\sqrt{33}+\sqrt{39}$

m $-4\sqrt{77}+4\sqrt{14}$

n $15\sqrt{11}+3\sqrt{55}$

o $-24\sqrt{14}+8\sqrt{6}$

p $15\sqrt{10}+20\sqrt{14}$

q $63\sqrt{5}$

r $11\sqrt{15}$

s $24\sqrt{77}-32\sqrt{55}$

10

a 60 b $\frac{12\sqrt{3}}{\sqrt{5}}$ c $\sqrt{5}+\sqrt{7}$ d $\frac{3\sqrt{3}}{\sqrt{10}}$

11

a $x=5999, y=-560$

b $x=5, y=-3$

12

a $42+(8+2m)\sqrt{13}+m^2$

b $m=-4$

Rationalise the denominator

13

a No b Yes c No

14

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a $\frac{\sqrt{11}}{11}$ **b** $\frac{\sqrt{22}-2\sqrt{11}}{11}$ **c** $\frac{7\sqrt{12}}{12}$ **d** $\frac{14\sqrt{143}}{117}$

15
a $\frac{\sqrt{3}}{3}$ **b** $7\sqrt{3}$ **c** $\frac{-9\left(\sqrt{5}+7\right)}{44}$ **d** $\frac{4\left(9\sqrt{11}+5\right)}{433}$
e $\frac{9+2\sqrt{14}}{5}$ **f** $\frac{\sqrt{10}-1}{9}$ **g** $4\left(\sqrt{10}+\sqrt{3}\right)$ **h** $\frac{5\sqrt{42}-4\sqrt{14}}{118}$

16
a $-\frac{2\sqrt{2}}{23}$ **b** $\frac{3\sqrt{5}-15\sqrt{2}}{20}$

17 No solution provided

18
a $\frac{\sqrt{3}-1}{2}$ **b** 16

19
a $19+8\sqrt{3}$ **b** $\frac{6-\sqrt{3}}{3}$

20 $\sqrt{1800}$

Applications

21 $12\sqrt{2}$ units

22 $3\sqrt{5}$ units

23
a $4\sqrt{7}\text{ m}$ **b** 3 m^2

24 $c=\sqrt{26}$ units

25 $9\sqrt{3}\text{ cm}$

26
a $\frac{\sqrt{5}+1}{2}$ **b** 1.618

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27 $4\sqrt{10} - 12$ m

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